

Analysis of open transmon system

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1 Transmon and harmonic oscillator

This section will derive and analyse behaviour of simple system consisting of a transmon modeled as Josephson junction and quantum harmonic oscillator.

1.1 The Josephson junction

The Josephson junction is made with two disconnected superconductors between which there is a layer of oxide. This setup allows Cooper pairs to tunnel throughout the junction.

To analyse the behaviour of Josephson junction we will only need two equations. One for the junction energy:

$$E = E_J \cos \varphi \quad (1)$$

And for the voltage across the junction:

$$V = \frac{\Phi_0}{2\pi} \frac{d\varphi}{dt} \quad (2)$$

1.2 Quantum LC circuit

We have a classical LC circuit with the Hamiltonian:

$$H_{LC} = \frac{Q^2}{2C} + \frac{\Phi^2}{2L} \quad (3)$$